

The UK Field Guide

RAG • Agentic Architecture • Multimodel input • Specie Classification • Vision

A vision-grounded wildlife assistant that turns a photograph into a cited, actionable answer.

August 2026

Project introduction

The problem, the audience, and the gap between identification and action

What the project is

Millions of people photograph wildlife every day...

Identification apps are good at answering 'what is this?', but almost none answer the question that matters next 'what should I do?'.

This project closes that gap for the UK, turning a single photo into an answer people can actually trust and act on, not just identified, but explained, sourced, and checked.

Goals

- Identify wildlife species with honest uncertainty displaying top-3 candidates and confidence.
- Ground every answer in retrieved, cited field-guide sources.
- Check regional status automatically and lead with safety warnings.

Who will use it

- Public users identifying a plant, insect, fungus, or bird on a walk.
- Local councils / conservation groups needing an invasive-status check.
- Agricultural bodies and land managers assessing risk.

Sources that feed the system

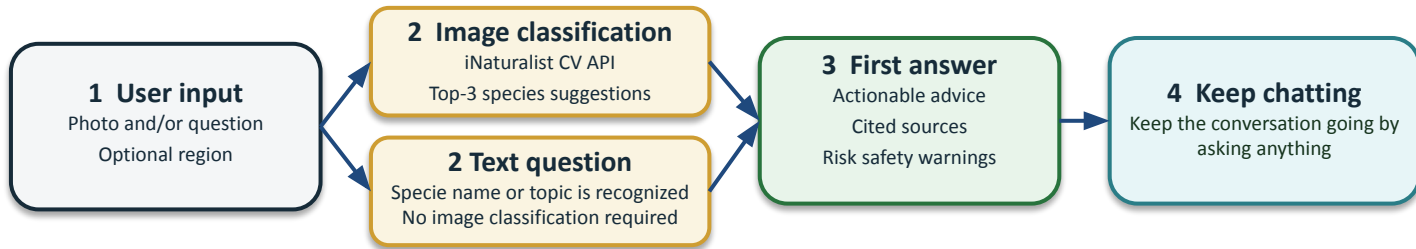
The system's knowledge layer aggregates data from iNaturalist (vision identification), Wikipedia (encyclopedic species information), Europe PMC/PubMed Central/arXiv/Open Library (scientific literature), and official UK Government invasive, protected-species, and conservation lists. In total, 362 documents across species accounts, regional status, books, and papers.

AI in the system

AI is embedded at three critical stages of the pipeline. First, a vision model identifies the species from a photo, returning ranked candidates with honest confidence scores rather than a single guess. Second, an LLM-powered agent grounds its answer in retrieved, cited field-guide sources, pulling only the most relevant passages rather than relying on its own memory. Third, a dedicated safety check reviews the answer before it reaches the user, flagging any protected or invasive-species concerns.

User journey

A minimal, outdoor-friendly interface built for four simple steps



1

User input

The user opens the app on their desktop or phone. They see a clean, outdoor-friendly interface with three input options: a photo upload field, a free-text question box, and a region selector. Either the photo or question is mandatory to trigger the app.

2

Image classification

The user selects a photo of a plant, insect, fungus, or bird. The system returns the top-3 specie names suggestions with confidence scores, displayed in a species bar above the chat. Only the top confidence score result will be passed into the final answer.

2

Text question

The user types a direct question (e.g., "Tell me about Japanese knotweed") without uploading an image. The system skips image classification. The system can recognize the specie name and topic.

3

First Answer

While the system reasons through the answer, the app streams a collapsible "thinking trace" so the user can watch each step unfold live. The system then combines everything it found, the field-guide details and the regional status check, into one clear, final answer. The output includes:

- Actionable advice tailored to the user's query.
- Inline citations linking back to specific sources for transparency.
- Safety warnings (e.g., invasive species alerts) which are enforced in code to never left to model judgement.

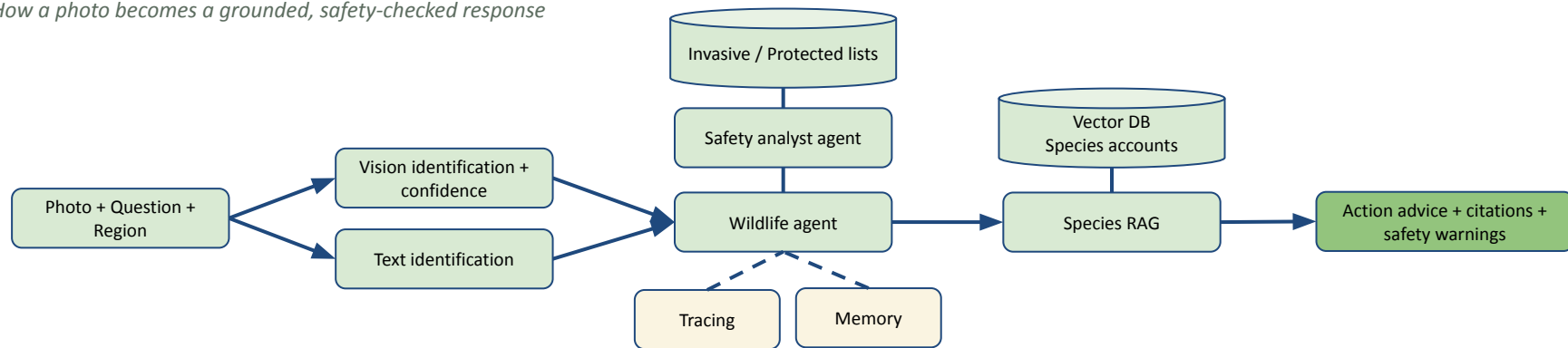
4

Keep chatting

The user keeps typing in the same chat, just like a normal chatbot conversation. The system understands follow-ups, detailed questions, a new species, or a comparison without needing another photo, and replies stays in the same thread.

Inside the pipeline

How a photo becomes a grounded, safety-checked response



Vision and Text Input

The process begins when the user submits a photograph or a question and an optional region. This multimodal input serves as the trigger for the entire workflow.

Specie Identification

The photograph is passed to a vision identification service, which returns the top-3 species candidates with confidence scores. This provides an honest, probabilistic starting point for the agent. The text question that is passed can identify the specie or subject.

Wildlife Agent

The core orchestrator is the Wildlife agent (Agent 1). This LLM-powered agent receives the vision results or a direct question, selects the right tools, builds a self-contained query, and begins its reasoning.

Safety analyst agent

The Wildlife agent consults a dedicated Safety Analyst agent (Agent 2,) who checks the list of invasive and protected species.

Species RAG

The Wildlife agent queries the Chroma vector database, which holds the curated field-guide corpus of species accounts. It pulls the 15 most relevant passages, then narrows them down to the 7 best matches, so the agent only works with the highest-quality, most focused information.

Final output

The LLM synthesises a grounded, RAG-based answer and streams it to the user token-by-token. The final response renders as clean Markdown, ending with a numbered source list for full transparency. Crucially, the system appends a mandatory safety warning, flagging any conservation, or invasive findings, enforced in code, not left to model judgement.

Tracing

Every action taken by the system is instrumented with tracing for observability. This enables real-time monitoring of latency, token usage, and tool calls, feeding into LangSmith for performance analysis.

Memory

The Wildlife agent maintains conversation memory and coordinates every downstream action, keeping the last 3 messages and folding earlier ones into a rolling summary.

Building the knowledge base

How the corpus is chunked, stored, and retrieved



Chunks

Before splitting, every document is whitespace-cleaned and hashed to drop duplicates (e.g. repeated book scans under different filenames), so chunking only ever runs on the cleaned, de-duplicated corpus.

The field-guide corpus is then split using LangChain's `CharacterTextSplitter` with a fixed-character strategy, the `separator=""`, which cuts purely by character count, not sentence or paragraph.

Chunk size is set to 1000 characters, balancing two competing needs: large enough to preserve meaningful context around a fact, but small enough for the embedding model to represent precisely and the reranker to score quickly. Chunk overlap is set to 100 characters, which guards against the fixed-character splitting strategy cutting a sentence or fact in half at an arbitrary boundary.

Key settings:

- Splitting strategy: fixed-characters split.
- Chunk size: 1000 characters.
- Chunk overlap: 100 characters.

Each chunk keeps its source filename, position in the document, and any citation metadata carried over from that source's record.

Embed & Vector database storage

Later each text chunks are converted into a numerical vector using OpenAI's embedding model, and stored in a local ChromaDB collection using Euclidean distance. This enables fast semantic similarity searches over the entire knowledge base.

Key settings:

- Embedding model: `text-embedding-3-small`.
- Persistence: stored in DB Species Accounts folder.
- Collection: `db_species_accounts`.

This layer is the foundation of the RAG pipeline. The embedding model determines how well the system understands the “meaning” of queries and documents. A more powerful model would improve recall but increase latency and storage costs. The local ChromaDB keeps everything self-contained and avoids network overhead for the retrieval step.

Retrieve & rerank

Based on the identified specie or user's question, the retrieval fetches the top-15 most similar documents from ChromaDB using embeddings. Then, a cross-encoder reranker re-scores these 15 candidates based on their semantic relevance to the actual query, filters out weak matches, and keeps only the top-7 most relevant passages for the agent.

Key settings:

- Initial retrieval (`DEFAULT_K`): 15 documents.
- Reranker model: `ms-marco-MultiBERT-L-12` (via `FlashrankRerank`).
- Score threshold: 0.7 (drops passages below this confidence).
- Final retrieval (`FINAL_k`): 7 documents.

This two-stage design is a classic cost/accuracy trade-off. Embedding-based retrieval is cheap but coarse. Reranking all 15 with a cross-encoder is more expensive, but by limiting that step to just 15 candidates, the pipeline stays fast while substantially improving precision. The 0.7 threshold aggressively prunes irrelevant chunks, ensuring that only high-confidence passages reach the LLM, reducing hallucinations at the risk of dropping useful but low-scoring context.

Two agents, one conversation

What each agent is responsible for, and how the app remembers what you've already discussed

Two agents share the work, one answers your questions by looking up facts in the field guide, and hands off to a dedicated safety expert for protected-species or invasiveness checks. The system keeps the conversation flowing smoothly by remembering only the most recent exchanges and a short summary of everything before that, so it never gets overwhelmed by a long chat.

Agent 1 — Wildlife Analyst

The agent is the primary user-facing assistant. It acts as a knowledgeable and safety-conscious wildlife expert. Its job is to answer the user's questions about wildlife, like species identification, ecology, threats, etc., by grounding every factual claim in the field-guide database.

This agent is instructed not to claim that an image-based identification is verified, to answer clearly and actionably, and to say explicitly when the retrieved context does not support an answer.

Capabilities Tools:

- `retrieve_documents` – fetches relevant passages from the vector store and must be called before answering any species-specific question.
- `summarise_topic` – produces up to 2000 words insightful account of a topic or species, ending with a concise summary.
- `compare_species` – provides a side-by-side comparison of two species using the field-guide data, like habitat, diet, behaviour, threats, status.
- `consult_safety_analyst` – delegates conservation and invasive-status checks to the dedicated safety expert, as this is agent-as-tool call into Agent.

Agent 2 — Safety Analyst

The agent is a specialist safety expert, never seen directly by the end user. It is consulted by Agent 1 whenever the conversation touches on legal protection, invasive status, or conservation status. Its role is to provide clear, cited, and actionable status checks.

This agent keeps its answers brief, in just a few sentences, because its output is used as input for Agent 1's final reply, not as the final answer to the user. It must also clearly state when the retrieved context does not support an answer and must not claim that an image-based identification is verified.

Capabilities Tools:

- `extract_conservation_status` – finds mentions of UK legal protection, etc., in the Invasive / Protected lists database.
- `check_invasive_status` – checks whether a species is flagged as invasive in a given region, first using a small reference table.

Conversation memory — Hybrid Approach

- **Window:** last 3 messages kept verbatim
- **Archiving trigger:** oldest message folded 1-by-1 into running summary
- **Cost:** summariser touches 1 message and summary, never full history
- **Token efficiency:** context size remains roughly flat regardless of conversation length
- **Context order:** agent receives System Prompt → Archived Summary → Recent 3 messages

✓ Pros

- **Bounded cost:** context stays near-flat as chats grow
- **Sharp recency:** exact wording kept for last 3 turns
- **Light footprint:** 1 message summarised at a time, low latency

✗ Cons

- **Lossy over time:** nuance and caveats get compressed away
- **Summary drift:** re-summarising can compound small errors
- **Fixed 3-msg window:** one static summary, as there is no lookback if a detail is asked about later

System settings comparison

Comparing today's setup against the changes we think will make it better

We want to know whether upgrading four core settings at once, the LLM, the reranker threshold, and the retrieval k-values, produces a meaningfully better assistant, not just a marginally different one.

The baseline and improved configurations are each run once against the same 30-question benchmark, under identical conditions, so both are scored on exactly the same questions. The improved configuration is expected to raise faithfulness and retrieval quality, but likely at the cost of higher latency and spend. The LLM is stronger, and the retrieval pool is wider, and this combination adds processing time and cost.

Because all four settings change together, though, this test can only show whether the bundle works as a whole, not which individual setting is responsible.

Setting	Baseline (current)	Improved (to test)	Why it matters
LLM model	gpt-4o-min (fallback model claude-sonnet-5)	gpt-5.6-Luna with max effort modes	This is the mode that both agents run. The baseline model is fast and cheap but can misstep on multi-tool reasoning. A newer and more powerful model, as the one selected in the improved settings, may reason better through multi-tool turns, understand complex queries, and reduce hallucination rates, even though at a higher cost and latency is expected.
Reranker threshold	Score Threshold = 0.7	Score Threshold = 0.6	The score threshold of 0.7 is the untuned default and may be discarding useful passages that fall just below the cutoff. Lowering it to 0.6 lets more context through, but a lower threshold means more but lower quality chunks.
Retrieval K-values	Initial retrieval = 15 Final retrieval = 7	Initial retrieval = 30 Final retrieval = 10	With the baseline setup, the agent only ever sees 7 passages out of 15 candidates, which is fine for simple questions, but potentially too narrow for harder ones. Widening the search to 30 candidates and keeping the top 10 gives the reranker more to choose from, at the cost of extra latency.

30 question-answer set

Samples of exchanges between the user and the model assistant in the four conversations, spanning topics used for benchmarking

American Bullfrog · Q1–8

Why is it called a “bullfrog”?

It gets its name from the sound the male makes during breeding season, which resembles a bull bellowing.

Is it considered invasive?

Yes — introduced to the Western US, South America, Western Europe, China, Japan, and Southeast Asia.

Small Emperor Moth · Q9–15

When do males fly and what do they look for?

Males fly rapidly by day from mid-April to late June, searching for sluggish, mostly nocturnal females.

What habitat does this moth prefer?

It's most often associated with heathland and moorland.

Swedish Whitebeam · Q16–22

How tall does it grow and what does its trunk look like?

It grows 10–20 m tall with a stout trunk, usually up to 60 cm in diameter, and grey bark.

What color do the leaves turn in autumn?

Dull yellowish to grey-brown.

Dogs vs Cats · Q23–30

How do they differ in communication styles?

Dogs are overt and expressive (barking, tail wagging); cats use subtler signals like purring and ear movement.

How do they differ in sensory abilities?

Dogs have a far superior sense of smell; cats have better night vision and hearing.

Note: All the question-Answer used for benchmark can be find in the appendix.

Observability results

Live performance metrics, tracked with LangSmith

The LangSmith results reveal a clear trade-off, with the improved settings maintaining the same reliability as the baseline while introducing significantly higher latency, token usage, and cost.

The corrected baseline re-run changes the earlier conclusion, with both configurations now at 0% error while the improved bundle is slower and more expensive across every remaining metric, with no measurable accuracy gain.

The results suggest that the next step is to test each setting in isolation to determine whether any individual improvement is worth the additional performance and cost overhead.

Metric	Baseline	Improved	Notes
Total tokens	236.7K	382.8K	Improved settings saw an increase of 146.1K tokens (+61.7%) due to using more model context and output overall.
LLM cost	\$0.0426	\$0.1641	Improved settings increased costs by 285% compared with the baseline, which means they are 3.85× more expensive to use.
Error rate	0%	0%	Both configs ran error-free.
P50 Latency	2.60s	7.06s	The median latency is 2.7× slower with the improved settings than with the baseline.
P99 Latency	13.67s	63.75s	The P99 latency is 4.7× slower with the improved settings, indicating a substantial tail-latency regression.
Trace count	119	121	There is a difference of only 2 traces, making the comparison broadly equivalent in volume.

Limitations and future improvements

Honest gaps in the current build, and where the roadmap goes next.

Limitations today

What the current build doesn't yet do well

Chunking is naive

Fixed-character splitting is applied across the corpus. This is a simple approach that cuts text at a set length rather than following natural breaks like sentences or paragraphs.

UK Government source gaps

Gov.uk invasive and conservation status documents that serve as source are very limited.

Retrieval quality unverified

Semantic ranking of retrieved chunks and the FlashRank reranker's real precision on this corpus haven't been verified by a specialist.

System settings tested simultaneously

System settings were compared only once, and all changes were applied at the same time.

Single region

Scoped to the UK only, and identification returns species suggestions with no seasonal or lookalike-species warning (e.g. "often confused with X").

Future improvements

Possible future fixes and the stretch goals from the project brief

Smarter chunking

Move from fixed-character splitting to a semantic or structure-aware splitter to keep species facts from being cut mid-sentence.

Complete the regional-status corpus

Finish ingesting the missing Gov.uk invasive and conservation-status documents, and grow the number of documents.

Re-run and report the evaluation set

Recommend re-running the evaluation with varied settings to validate consistency across multiple comparison passes.

Offline-friendly mode

Explore a locally run vision model for identification, reducing dependence on the iNaturalist API.

Appendix - 30 question-answer set (1/4)

Four real conversations, used to benchmark answer quality

Conversation 1: American Bullfrog (*Lithobates catesbeianus*) — Questions 1–8

1:

Question: What is the scientific name of the American bullfrog and where is it native to?

Answer: *Lithobates catesbeianus* (sometimes *Rana catesbeiana*). It is native to eastern North America.

2:

Question: Why is it called a "bullfrog"?

Answer: The male's call during breeding season sounds like a bull bellowing.

3:

Question: What types of habitats does the American bullfrog typically inhabit?

Answer: Large permanent water bodies like swamps, ponds, and lakes, as well as manmade habitats like pools, canals, ditches, and culverts.

4:

Question: Is the American bullfrog considered an invasive species? If so, where?

Answer: Yes. It has been introduced into the Western US, South America, Western Europe, China, Japan, and Southeast Asia.

5:

Question: Why are bullfrogs considered invasive and harmful to local ecosystems?

Answer: Due to their voracious appetite and high egg production, they negatively affect native amphibians, insects, and other fauna.

6:

Question: What does the American bullfrog look like in terms of coloration?

Answer: The dorsal surface is olive-green (plain or mottled with grayish-brown), and the ventral surface is off-white blotched with yellow.

7:

Question: For what purposes are bullfrogs used by humans?

Answer: They are eaten (especially in the southern US), used for dissection, kept as pets (albino variants), and their tadpoles are sold for ponds.

8:

Question: Who is the species named after?

Answer: English naturalist Mark Catesby (*catesbeianus* / *catesbeiana*).

Appendix - 30 question-answer set (2/4)

Four real conversations, used to benchmark answer quality

Conversation 2: Small Emperor Moth (*Saturnia pavonia*) — Questions 9–15

9:
Question: What is the scientific name of the small emperor moth and who first described it?
Answer: *Saturnia pavonia*, first described by Carl Linnaeus in 1758.

10:
Question: What family does it belong to and where can it be found?
Answer: Family Saturniidae. It occurs throughout the Palearctic region and is the only member of its family found in the British Isles.

11:
Question: What are the main differences in appearance between males and females?
Answer: Males are smaller (~60 mm wingspan) with brown/white forewings and bold eyespots. Females are larger (~80 mm), less brightly coloured (grey/white), but have similar eyespots.

12:
Question: When do males fly and what do they look for?
Answer: Males fly rapidly during the day from mid-April to late June, looking for sluggish females that usually fly only at night.

13:
Question: What habitats does this moth prefer?
Answer: Heathland and moorland.

14:
Question: What does the caterpillar look like and what does it eat?
Answer: Initially black and orange, later green with black rings and yellow/red spots. Its commonest food plant is heather (*Calluna*), but it feeds on many other plants.

15:
Question: What is a common incorrect genus name sometimes used for this species?
Answer: *Pavonia*.

Appendix - 30 question-answer set (3/4)

Four real conversations, used to benchmark answer quality

Conversation 3: Swedish Whitebeam (*Scandosorbus intermedia*) — Questions 16–22

16:

Question: What is the scientific name of the Swedish whitebeam and where is it found?

Answer: *Scandosorbus intermedia* (formerly *Sorbus intermedia*). Found in southern Sweden, with scattered occurrences in Estonia, Latvia, Bornholm, SW Finland, and northern Poland.

17:

Question: What family does it belong to?

Answer: Family Rosaceae.

18:

Question: How tall does it grow and what does its trunk look like?

Answer: 10–20 m tall, with a stout trunk up to 60 cm in diameter (sometimes 2–3 m) and grey bark.

19:

Question: What shape is the crown?

Answer: Dome-shaped, with stout horizontal branches.

20:

Question: What do the leaves look like?

Answer: Green above, densely hairy with pale grey-white hairs beneath, 7–12 cm long, with 4–7 oval lobes per side, rounded at the apex, and finely serrated margins.

21:

Question: What color do the leaves turn in autumn?

Answer: Dull yellowish to grey-brown.

22:

Question: What do the flowers look like?

Answer: 15–20 mm in diameter, with five white petals and 20 yellowish-white stamens, produced in corymbs 8–12 cm across.

Appendix - 30 question-answer set (4/4)

Four real conversations, used to benchmark answer quality

Conversation 4: Dogs vs Cats (general comparison) — Questions 23–30

23:

Question: What are the wild relatives of dogs and cats?

Answer: Dogs' relatives include wolves, coyotes, and jackals; cats' relatives include lions, tigers, and leopards.

24:

Question: How do dogs and cats differ in social structure and relationship with humans?

Answer: Dogs are obedient followers adapted to pack structures where humans can be pack leaders. For cats, humans must adopt a more modest position.

25:

Question: What are the main categories of comparison typically covered in books about cats vs. dogs?

Answer: Senses, diet, communication, and behaviour.

26:

Question: What is the evolutionary and behavioural approach to keeping them together?

Answer: Some books examine biological and behavioural differences to devise strategies enabling both species to live in harmony in the same household.

27:

Question: What is a well-known children's book that contrasts their behaviours?

Answer: Cats Do, Dogs Don't.

28:

Question: How do they differ in sensory abilities?

Answer: Dogs have a far superior sense of smell, while cats have better night vision and hearing.

29:

Question: How do they differ in communication styles?

Answer: Dogs are more overt and expressive (barking, tail wagging, body postures); cats use more subtle signals (purring, tail positioning, ear movements).

30:

Question: What is the key behavioural difference in training?

Answer: Dogs are generally more trainable and cats are more independent and less responsive to traditional training.